symbols following the arrow show what has become of this atom as a result of the emission of the alpha particle. In such a formula, the sum total of the superscripts must be equal on both sides of the arrow; in this particular case, 226 = 222 + 4. This follows from the law of the conservation of mass. The same rule applies to the subscripts; in this case, 88 = 86 + 2. This follows from the law of the conservation of electric charges. Analogous formulae are used for processes in which beta radiation occurs.

All—or nearly all—of the alpha particles emitted by a homogeneous radioactive substance have exactly the same range. This is evident in the cloud chamber photograph shown in Figure 3 (*a*). It shows the successive decay of two radioactive elements; therefore, we see two groups of alpha rays. The range in air of the alpha particles of different radioactive substances is between 1 and 9 centimetres, and the range is evidently dependent on the energy with which the particles emerge from the nucleus. The greater the energy, the longer is the range.

The various radioactive substances show great differences in their respective speeds of transmutation. Some of these substances are very short-lived, whilst others last very long and show no noticeable lessening in radioactivity over long periods of time. Obviously, for the atoms of every radioactive substance there exists a probability, capable of being expressed numerically, of their radioactive decay. The reciprocal of this probability is the *average life* of the substance. The decay probability, and hence also the average life, is independent of the number of atoms already decayed. This means that the same percentage of the number of atoms still intact will decay per unit time. This law is expressed by the following equation:

$$dN = -\lambda N\, dt$$

with the following solution for N:

$$N = N_0 e^{-\lambda t}$$

where N_0 is the number of the intact atoms present at the time $t = 0$, N is their number at the time t, e is the base of natural logarithms, and λ is the decay